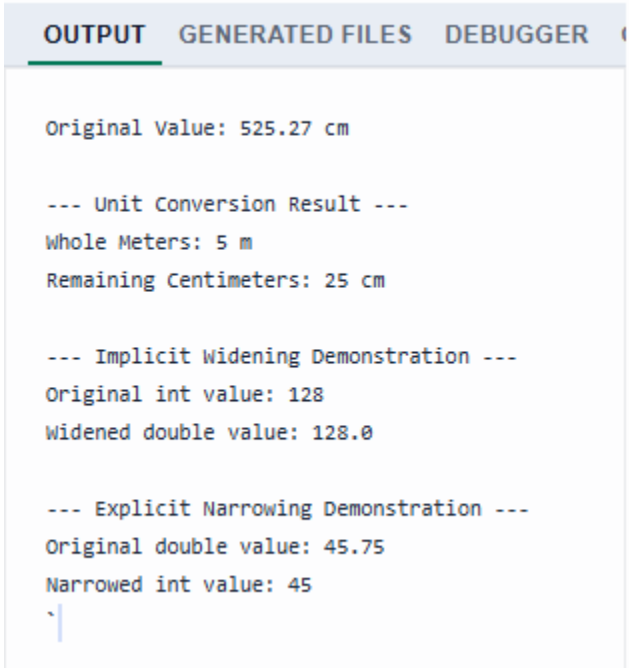
```
public class UnitCastingConverter {
    public static void main(String[] args) {

        // 1. Declare a double value representing centimeters
        double totalCentimeters = 525.27;
        System.out.println("Original Value: " + totalCentimeters + " cm\n");

        // 2. Compute whole meters and remaining centimeters using explicit narrowing casts
        // Casting a double to an int truncates the decimal part, leaving only the whole number.
        int totalMeters = (int) (totalCentimeters / 100);
        int remainingCentimeters = (int) (totalCentimeters % 100);

        System.out.println("--- Unit Conversion Result ---");
        System.out.println("Whole Meters: " + totalMeters + " m");
        System.out.println("Remaining Centimeters: " + remainingCentimeters + " cm");
        System.out.println();

        // 3. Separate Demonstration: Implicit Widening
        int initialCentimeters = 128;
        // IMPLICIT WIDENING: An 'int' (32-bit) automatically scales up to a 'double' (64-bit).
        // No explicit cast is needed because a double has a larger range and precision capacity
than an int.
        // There is zero risk of data loss. Java handles this automatically.
        double widenedCentimeters = initialCentimeters;

        System.out.println("--- Implicit Widening Demonstration ---");
        System.out.println("Original int value: " + initialCentimeters);
        System.out.println("Widened double value: " + widenedCentimeters);
        System.out.println();

        // 4. Separate Demonstration: Explicit Narrowing
        double precisionMeasurement = 45.75;
        // EXPLICIT NARROWING: Moving from a 'double' (64-bit) down to an 'int' (32-bit).
        // An explicit '(int)' cast is required because we are forcing a larger, more precise type into a
smaller type.
        // FIXED COMMENT: The entire fractional decimal portion (.75) is permanently truncated
(chopped off),
        // NOT rounded. The variable 'narrowedMeasurement' only retains the whole number.
        int narrowedMeasurement = (int) precisionMeasurement;

        System.out.println("--- Explicit Narrowing Demonstration ---");
        System.out.println("Original double value: " + precisionMeasurement);
        System.out.println("Narrowed int value: " + narrowedMeasurement);
    }
```

}


OUTPUT
GENERATED FILES
DEBUGGER
Original Value: 525.27 cm
--- Unit Conversion Result ---
Whole Meters: 5 m
Remaining Centimeters: 25 cm
--- Implicit Widening Demonstration ---
Original int value: 128
Widened double value: 128.0
--- Explicit Narrowing Demonstration ---
Original double value: 45.75
Narrowed int value: 45